

CIRA Technical Implementation & Progress Report

Comprehensive Technical Documentation of Recent Enhancements, Bug Fixes & Architecture Updates

Date: September 13, 2026	System Status: All Services Active
Backend Core: http://localhost:3000	Frontend Client: http://localhost:5175
Database Layer: Neon PostgreSQL + MongoDB Atlas	Environment: Production / Cloud Direct

Executive Summary

This document synthesizes the technical enhancements, bug fixes, and data pipeline optimizations implemented across the CIRA platform. Key updates include direct cloud database integration (Neon PostgreSQL + MongoDB Atlas), faculty topic-based question authoring, auto-grading persistence, actionable insights calculation for student weak topics, and clock-synced exam portal timer enforcement.

1. Core Infrastructure & Database Connectivity

- **Direct Cloud Connection:** Backend core connects directly to external Neon PostgreSQL and MongoDB Atlas clusters without relying on local Docker databases.
- **Environment Configuration:** Preserved active credentials in backend-core/.env to maintain continuous runtime stability.
- **Active Microservices:** Verified healthy execution of both backend API gateway/core (port 3000) and Vite frontend web application (port 5175).

2. Topic-Based Question Authoring & DOCX Parser

- **Prisma Schema & Controller Support:** Enhanced Question model and addQuestions handler in quiz.controller.ts to record question topics in PostgreSQL.
- **DOCX Parser Upgrade:** Enhanced uploadDocxParser to auto-detect Topic: headers when importing bulk question sets.
- **Faculty UI Expansion:** Updated QuizManagement.tsx with full topicSections state support and a dedicated '+ Add Another Topic Section' workflow.

3. Weak Topic Calculation & Actionable Insights Engine

Problem Identified: Previously, submitExam calculated objective scores in memory but did not record marksAwarded per response in PostgreSQL. Consequently, getQuizAnalytics returned empty topic averages ([]), displaying 'No weak topics detected' even for failing student attempts.

Technical Resolution:

- **Submission Persistence:** Updated submitExam in student-exam.controller.ts to iterate over all quiz questions and write individual marksAwarded to QuizResponse records upon attempt submission.
- **Analytics Reporting:** Updated getQuizAnalytics in faculty-reports.controller.ts to group responses by question.topic and calculate topic score averages (with on-the-fly evaluation fallback for legacy attempts).
- **Actionable Insights UI:** Topic averages below 65% automatically surface under Actionable Insights in StudentReports.tsx with immediate options to generate remedial assignments.

4. Exam Portal Countdown Timer & Security Enforcement

Problem Identified: ExamInterface.tsx initialized its timer state to 3600s prior to API load, causing a visible clock jump down to remaining time upon fetch, along with state drift across background tabs.

Technical Resolution:

- **Absolute Timestamp Diffing:** Refactored ExamInterface.tsx to compute an absolute target end timestamp: $\text{endTimeMs} = \text{startTimeMs} + (\text{durationMinutes} * 60 * 1000)$.
- **Clock Synchronization:** The 1-second interval loop derives remaining seconds as $\text{timeLeft} = \max(0, \text{floor}((\text{endTimeMs} - \text{Date.now()}) / 1000))$.
- **Instant Auto-Submit:** Guarantees drift-free countdown accuracy and triggers instant auto-submission when $\text{timeLeft} \leq 0$.

5. Score Percentage & Performance Band Correctness

Problem Identified: Default quiz totalMarks of 100 caused a 7 out of 10 score ($\text{rawScore} = 7$) to evaluate as $(7 / 100) * 100 = 7\%$, incorrectly placing the student in the 'Needs support' band ($< 60\%$).

Technical Resolution:

- **Dynamic Question Summation:** Updated getQuizAnalytics and getPerformanceBands in faculty-reports.controller.ts to calculate total marks from the sum of question marks (10 total marks).
- **Percentage & Band Accuracy:** A score of 7 out of 10 now evaluates to **70%** and correctly assigns the student to the **Average** band (60%–79%).
- **Automatic DB Synchronization:** Updated addQuestions in quiz.controller.ts to update quiz.totalMarks in PostgreSQL whenever questions are saved.

6. System Verification Matrix

Component / Area	Target Path / Module	Verification Status
Backend Core	backend-core/src/controllers/*	Passed (tsc code 0)
Frontend Web App	frontend-web/src/*	Passed (tsc code 0)
Cloud Database	Neon PostgreSQL + MongoDB Atlas	Connected & Syncing
Exam Portal	ExamInterface.tsx	Clock Synced
Faculty Reports	StudentReports.tsx	Insights Enabled